MILITARY SPECIFICATION

ROTARY SWITCH AND HARNESS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1 SCOPE

1.1 This specification covers the procurement requirements for one type of rotary arming switch and harness, hereinafter referred to as "assembly", for use as part of the arming system on the Mark 56 Mods 0 and 1 Rocket Motors.

2 APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

Military

MIL-G-45204	Gold Plating, Electrodeposited

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

DRAWING

Naval Ordnance Systems Command

2806551	Rotary Switch and Harness

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3 REQUIREMENTS

3.1 First article. First articles shall be manufactured using the method proposed for production. These first articles will be tested to determine if the proposed production method yields an item which meets the requirements of this specification. The number of first articles shall be as specified in the procurement document invoking this specification.

3.2 Design and construction. The assembly shall consist of a rotary switch, actuated with a removable control rod, and associated wiring and hardware conforming to Drawing 2806551, hereinafter referred to as "the switch". The switch shall function properly when mounted in any position.

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3.3 Materials. Unless otherwise specified, all metal parts shall be of corrosion-resistant material or shall be suitably plated to resist corrosion. Solder terminals shall be treated to facilitate soldering; coatings such as hot solder, hot tin dip, or gold plating are acceptable. Contacts shall be gold plated in accordance with MIL-G-45204, Type II, Grade C or D, Class 3 (99 percent gold minimum).

3.4 Cable integrity. The switch cables shall withstand without damage an applied force of 15 pounds (67N) for one minute. There shall be no visible signs of cracks, breaks, or distortions at the area where the cables attach to the switch. See 6.3.

3.5 Operating life. The switch shall be capable of withstanding without damage 300 operating cycles, consisting of rotation from the open position stop to the closed position stop and back to the open position stop.

3.6 Environmental.

3.6.1 Temperature. The switch shall withstand exposure to a storage temperature range of minus 10° F to plus 120° F (-23° to 49° C). It shall operate and function properly between plus 45° F and plus 95° F (7° to 35° C).

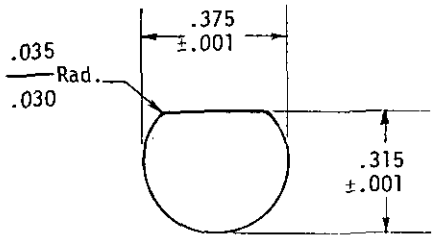
3.6.2 Vibration.

3.6.2.1 Open position. When vibrated at 3.5 units of gravity (g) (34 m/s^2) at 20 hertz (Hz), and 0.5 g (5 m/s^2) at 300 Hz, the circuit between the output connector contacts and ground shall not open for a period longer than 10 microseconds (µs).

3.6.2.2 Closed position. When vibrated at 3.5 g (34 m/s^2) at 20 Hz, the circuit between the input terminal and the output connector contacts shall not open for a period longer than 100 µs.

3.7 Torque. The torque required to rotate the switch rotor from stop-to-stop shall not exceed 2 inch-pounds (0.23/N·m).

3.8 Angular lash. The total angular lash between the switch rotor and a control rod having the dimensions shown in Figure 1, when the rod is rotated through its full clockwise and counterclockwise clearance, shall be a maximum of 7°.



IN	MM
.001	0.025
.030	0.76
.035	0.89
.315	8.00
.375	9.53

Note: Dimensions shown in inches.

Figure 1.
Cross-section of control rod used for testing.

3.9 Electrical.

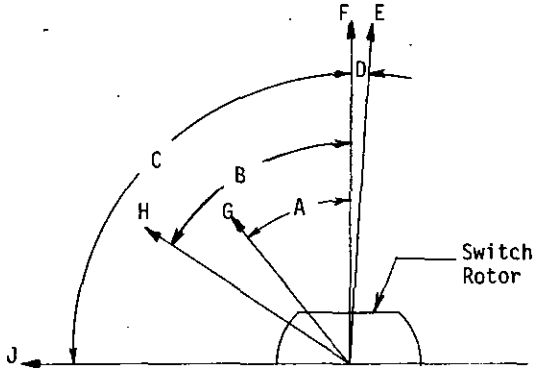
3.9.1 Resistance.

3.9.1.1 Open position. While the switch rotor is rotated from the open position stop to position one (see Figure 2), the resistance between the connector output contacts "A" and "B" and the

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ground wire terminal shall not exceed 0.07 ohm. Simultaneously, there shall be an open circuit between the harness input terminals "A", "B", "C", and "D" and the connector output contacts.

3.9.1.2 Closed position stop. While the switch rotor is rotated from position two to the closed position stop (see Figure 2), the resistance between the harness input terminals "A" and "C", "B" and "D" and the connector output contacts "A" and "B", respectively, shall not exceed 0.25 ohm. Simultaneously, there shall be an open circuit between the connector output contacts and the ground wire terminal.



A	37° - 46°
B	51° - 57°
C	85° - 90°
D	5° - 10°
E	Open Position Stop
F	Zero Position
G	Position One
H	Position Two
J	Closed Position Stop

Zero position is established when the flat of rotor is perpendicular to the "A"-plane shown on Drawing 2806551.

Figure 2
Switch positions and tolerances.

3.9.1.3 Shield. The electrical resistance between the connector ground contact "G" and the cable shielding at the harness terminals shall not exceed 0.25 ohm.

3.9.1.4 Insulation resistance. Insulation resistance shall be not less than 1000 megohms at 500 volts direct current (vdc) between circuits; and between the circuits and ground.

3.9.2 Electrical rating. With the switch in the closed position, the assembly shall carry no less than 10 amperes dc for no less than 100 milliseconds when 28 vdc power, rated at no less than 10 amperes, is applied across contacts "A" and "B" of the output connector with the input terminals shorted together.

3.10 Workmanship. Components and the assembly shall show no evidence of cracks, breaks, dents, burrs, abrasions, foreign material, or other indication of poor workmanship that could adversely affect their use, and shall be uniform in quality.

4 QUALITY ASSURANCE PROVISION

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (4.4)

b. Quality conformance inspection (4.5)

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4.3 Test equipment. Inspections of the arming switch and harness will require test equipment listed below. Unless otherwise specified, all test equipment parameters and measurements shall be accurate within 10 percent or 10° of true value, as appropriate.

a. Regulated Power Supply, 0-40 VDC, Hewlett-Packard Model 6267B, or equivalent.

b. Igniter Circuit Tester, 0-30 ohm, maximum current 5 ma, 1 percent accuracy full scale, Alinco Model 101-5AF, or equivalent.

c. Ammeter, DC, Weston Model 931, or equivalent.

d. Megohmmeter 50/500 VDC range, 3% accuracy, General Radio Model 1862B, or equivalent.

e. Torque Wrench, Torqometer Model TQS-205, or equivalent.

f. Temperature Control Unit, Conrad/Missimer Model FB1.5, or equivalent.

g. Vibration table capable of 20 Hz at 3.5 g (34 m/s^2) and 300 Hz at 0.5 g (5 m/s^2) vibration of the arming switch.

h. Chatter detector capable of measuring contact chatter of less than 10 µs duration.

4.4 First article inspection. Unless otherwise specified in the contract or order, six assemblies shall be subjected to first article inspection. The inspections shall be performed in the order listed in Table I.

4.5 Quality conformance inspection.

4.5.1 Sampling. Unless otherwise specified, a sample shall be drawn from each lot in accordance with MIL-STD-105, using Level II inspection with an acceptable quality level (AQL) of 1.0 percent.

4.5.2 Lot size. A lot shall consist of all assemblies from an identifiable production period, from one manufacturer, using switches from one similarly identifiable lot.

4.5.3 Examination and tests. Quality conformance inspection shall be as listed in Table I. Inspection shall apply to 100% of each lot, except as indicated in Table I.

Table I - Examinations and Tests

Test or Examination	Requirements Paragraph	Method Paragraph	Order of testing	
			First Article	Quality Conformance
Inspection	3.2,3.3,3.10,5	4.6.1	1	1
Cable integrity	3.4	4.6.2	2	2*
Angular lash	3.8	4.6.4	3	3*
Torque	3.7	4.6.5	4	4*
Resistance	3.9.1	4.6.6	5	5
Operating life	3.5	4.6.3	6	
Conditioning	3.6.1	4.6.7.1	7	
Low temperature	3.6.1	4.6.7.2	8	
Resistance	3.9.1	4.6.6	9	
Torque	3.7	4.6.5	10	
High temperature	3.6.1	4.6.7.3	11	
Resistance	3.9.1	4.6.6	12	
Torque	3.7	4.6.5	13	
Vibration	3.6.2	4.6.8	14	
Resistance	3.9.1	4.6.6	15	
Torque	3.7	4.6.5	16	
Electrical rating	3.9.2	4.6.9	17	
*To be performed only on the samples of 4.5.1.				

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4.6 Test Methods.

4.6.1 Dimensional, material, and workmanship. The assembly shall be inspected for conformance to design, construction, materials, workmanship, and delivery preparation requirements.

4.6.2 Cable integrity. With the switch securely held in a suitable fixture, the cable shall be pull tested at an applied force of not less than 15 pounds (67N) for not less than one minute.

4.6.3 Operating life. The switch shall be cycled 300 times. Time to rotate through one cycle shall be 0.30 to 0.50 second.

4.6.4 Angular lash. With a control rod having the dimensions of Figure 1 inserted in the switch rotor, the total counterclockwise and clockwise angular lash shall be measured within one degree of arc.

4.6.5 Torque. The maximum torque required to rotate the switch rotor from open stop position to closed stop position and back to open stop position shall be measured using the device of 4.3e.

4.6.6 Resistance tests.

4.6.6.1 Contacts and terminals. While the switch rotor is rotated from the open stop position to the closed stop position, the electrical resistance shall be continuously measured with the device of 4.3b between the following contacts and terminals:

a. Connector contact "A" and ground "G"
b. Connector contact "B" and ground "G"
c. Connector contact "A" and terminal "A"
d. Connector contact "A" and terminal "C"
e. Connector contact "B" and terminal "B"
f. Connector contact "B" and terminal "D"

4.6.6.2 Shield. The electrical resistance shall be measured between connector contact "C" and the cable shielding at the harness terminals "C" and "D", using the device of 4.3b.

4.6.6.3 Insulation resistance. The assembly shall be subjected to the test specified in MIL-STD-202, Method 302, Test Condition B, using the device of 4.3d. With the switch in the open position, the tests shall be made between terminals "A" and "C" and "B" and "D" and connector contacts "A" and "B", and between terminals "A" and "C" and "B" and "D" and ground "G". With the switch in the closed position, the tests shall be made between circuit connector contact "A" and terminals "A" and "C", between circuit connector contact "B" and terminals "B" and "D", and between the two preceding circuits and ground "G". Tests shall also be made between connector contact "C" and connector contacts "A" and "B", between connector contact "C" and terminals "A" and "C" and "B" and "D", and between connector contact "C" and ground "G".

4.6.7 Temperature. Temperature tests shall utilize the device of 4.3.f.

4.6.7.1 Conditioning. The assembly shall be temperature conditioned at minus 10° ± 5°F (-23° ± 3°C) for 24 hours, minimum, and allowed to return to room temperature. The assembly shall then be temperature conditioned at plus 120° ± 5°F (49° ± 3°C) for 24 hours, minimum, and allowed to return to room temperature before subjecting the assembly to the tests of 4.6.7.2 and 4.6.7.3.

4.6.7.2 Low temperature. The assembly shall be placed in a cold chamber and conditioned for four hours, minimum, at plus 45° ± 5°F (7° ± 3°C). The switch shall be subjected to the tests of 4.6.5 and 4.6.6 while at this temperature.

4.6.7.3 High temperature. The assembly shall be conditioned and tested as specified in 4.6.7.2; except the temperature shall be plus 95° ± 5°F (35° ± 3°C).

4.6.8 Vibration. The assembly shall be mounted on a rigid, metal panel. The mounting fixture shall be free from resonance at the frequencies of the tests. The rotor of the switch shall be locked against rotation in the specified position by means of a shaft through the hub of the

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rotor. The shaft shall have the dimensions shown in Figure 1 for its length inside the hub of the rotor. The harness terminals and connector shall be rigidly attached to the panel with enough slack in the wire leads between terminals to prevent damage to the wires or interference with vibration during the testing.

4.6.8.1 Open position. With the switch rotor in the zero position, the harness shall be vibrated at 3.5 g (34 m/s^2), using the device of 4.3 g, for no less than 36 hours at 20 Hz, and at 0.5 g (5 m/s^2) for no less than 36 hours at 300 Hz. The harness shall be vibrated for an equal time in each of three mutually perpendicular axes. One axis of vibration shall be parallel to the axis of the rotor of the switch. Switch contact chatter of the output-to-ground circuits shall be monitored continuously during vibration using the device of 4.3h.

4.6.8.2 Closed position. With the switch rotor 70° from the zero position, the harness shall be vibrated for no less than 20 minutes in each of the three mutually perpendicular axes of 4.6.8.1, at 3.5 g (34 m/s^2) and 20 Hz using the device of 4.3 g. Switch contact chatter of the two circuits (connector contact "A" and terminals "A" and "C", and connector contact "B" and terminals "B" and "D") shall be monitored continuously during vibration, using the device of 4.3h.

4.6.9 Electrical rating . (This is a destructive test.) With the switch in the closed position and the input terminals short-circuited together, 28 VDC shall be applied across the output connector contacts "A" and "B", using the device of 4.3a. The DC current through the switch shall be monitored for no less than 100 milliseconds, using the device of 4.3c.

5 PREPARATION FOR DELIVERY

5.1 Packing. Unless otherwise specified, switches shall be packed in clean, dry containers. The packing container shall be so constructed as to ensure acceptance by common or other carrier for safe transportation, at the lowest rate, to the place of delivery specified by the contract or order.

5.2 Marking. Containers shall be marked in accordance with MIL-STD-129. Marking shall include, but not be limited to, the following information:

a. Number and revision letter of this specification
b. Manufacturer's name
c. Lot number
d. Switch lot number
e. Number of assemblies in the lot
f. Purchase order number

6 NOTES

6.1 Intended use. The arming switch and harness is intended for use in a rocket motor as part of the arming system.

6.2 Ordering data. Procurement documents should specify, but not be limited to, the following:

a. Place of inspection
b. Place of delivery
c. Number and revision letter of this specification
d. Inspection records to be transmitted and method of transmittal to procuring activity.

6.3 Metrication. This document has been annotated parenthetically with metric equivalents of quantities not expressed in the International System of Units. The metric units are in accordance with ASTM E380-72 as adopted by the DoD on 29 November 1973, and are included for the purpose of encouraging metrication. Parenthetical quantities are intended for information only, as they are usually not exact equivalents but have been rounded off within tolerances specified herein.

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Equivalent quantities of particular interest herein include:

1.000 pound (force)	= 4.482 Newtons (N)
1.000 gravity	= 9.81 metre/second^2 (m/s^2)
1.000 inch-pound (torque)	= 0.1130 newton-metre (N·m)

6.4 Technical cognizance for the rotary switch and harness described herein is located at the Naval Ordnance Station, Code 5721, Indian Head, MD 20640.

Custodian:
NAVY - OS

Preparing Activity:
NAVY - OS
Project Number:
1338-N030

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Naval Ordnance Station
Indian Head, MD 20640

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-R-82642(OS) "ROTARY SWITCH AND HARNESS"

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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